

NITRA, Slovakia

WMO: 118550

Lat: 48.283N

Lon: 18.133E

Elev: 136

StdP: 99.70

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.5	-8.7	-15.1	1.0	-9.7	-12.3	1.3	-6.6	13.8	5.5	12.5	3.0	1.8	320	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	32.9	21.0	31.0	20.3	29.3	19.6	22.0	30.1	21.2	29.0	20.4	27.7	3.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.3	25.3	18.5	13.6	24.4	17.8	13.0	23.5	64.9	30.1	62.0	29.1	59.3	28.0	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	10.6	9.4	-14.9	35.1	4.0	1.7	-17.8	36.3	-20.1	37.3	-22.4	38.3	-25.3	39.5		
(5)			-15.2	23.3	3.9	0.8	-18.1	23.8	-20.4	24.3	-22.6	24.8	-25.4	25.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	-0.8	1.5	5.7	11.4	15.7	19.2	21.3	20.9	16.1	10.9	5.8	0.5
	DBStd	8.64	4.41	4.28	3.97	4.08	3.52	3.45	3.33	3.34	3.50	4.03	4.11	4.34
	HDD10.0	1217	335	239	142	32	2	0	0	0	1	39	134	293
	HDD18.3	3059	593	472	392	210	96	30	11	13	83	232	375	551
	CDD10.0	1486	0	1	8	73	179	276	349	339	185	66	9	0
	CDD18.3	286	0	0	0	2	14	56	102	93	18	1	0	0
	CDH23.3	3181	0	0	0	30	192	633	1152	993	176	6	0	0
	CDH26.7	1122	0	0	0	3	32	205	463	384	34	0	0	0
Wind	WSAvg	3.9	4.0	4.2	4.5	4.2	3.9	3.6	3.6	3.3	3.5	3.7	4.4	4.0
Precipitation	PrecAvg	614	37	36	36	43	65	65	71	56	56	50	51	46
	PrecMax	943	66	97	97	103	172	142	154	111	162	138	115	129
	PrecMin	383	2	1	3	1	20	14	3	3	10	5	4	9
	PrecStd	110	18	22	26	25	36	31	38	33	36	31	26	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	14.1	19.9	26.5	29.4	33.2	35.0	35.5	29.9	24.0	17.1	11.0
		MCWB	8.2	8.9	11.4	15.7	18.7	21.5	21.5	21.1	19.0	16.6	13.2	8.7
	2%	DB	8.2	11.3	17.2	23.7	27.3	31.1	33.0	32.9	27.3	21.5	14.8	8.9
		MCWB	6.4	7.8	10.3	14.4	18.0	20.9	21.1	20.8	18.1	15.5	11.8	7.5
	5%	DB	6.5	9.2	14.7	21.6	25.6	29.1	31.1	30.7	25.3	19.6	13.2	7.4
		MCWB	5.0	6.7	9.2	13.3	17.2	19.9	20.1	20.2	17.2	14.7	10.5	6.1
	10%	DB	5.0	7.6	12.5	19.3	23.6	27.1	29.3	28.7	23.2	17.4	11.7	6.0
		MCWB	3.8	5.6	8.1	12.3	16.1	18.9	19.6	19.3	16.3	13.6	9.7	4.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	9.8	12.4	16.4	20.4	23.4	22.7	22.8	20.2	17.5	13.6	9.1
		MCDB	10.8	13.2	17.7	24.6	27.0	30.8	31.8	31.1	27.2	22.1	16.3	10.6
	2%	WB	6.7	8.2	11.0	15.0	18.9	21.8	21.8	21.7	19.1	16.2	12.2	7.6
		MCDB	8.0	10.7	15.8	22.2	25.7	29.0	30.4	29.9	25.2	20.4	14.5	8.7
	5%	WB	5.1	6.9	9.8	13.9	17.8	20.8	21.1	20.9	18.1	15.1	11.0	6.2
		MCDB	6.2	8.9	14.0	20.2	24.0	27.6	29.2	29.0	23.3	18.7	12.8	7.2
	10%	WB	3.8	5.7	8.6	12.8	16.8	19.7	20.3	20.0	17.1	14.0	9.8	4.9
		MCDB	4.8	7.5	11.9	18.3	22.3	25.8	27.8	27.2	22.1	17.0	11.5	5.7
Mean Daily Temperature Range	5% DB	MDBR	5.5	7.1	9.5	11.4	11.5	11.9	12.4	12.2	10.7	9.1	6.0	4.8
		MCDBR	6.7	9.7	13.6	14.9	14.9	15.0	15.5	15.3	14.2	12.0	8.1	5.7
		MCWBR	5.2	6.8	8.0	7.4	7.1	6.5	5.7	5.7	6.6	6.9	5.5	4.6
	5% WB	MCDBR	6.3	8.8	11.9	13.3	13.3	13.3	14.0	13.6	11.9	10.5	7.1	5.3
		MCWBR	5.1	6.4	7.4	6.9	6.8	6.4	5.6	5.6	5.8	6.3	5.1	4.4
Clear-Sky Solar Irradiance	taub	0.324	0.347	0.384	0.431	0.427	0.427	0.438	0.437	0.410	0.381	0.358	0.323	
	taud	2.346	2.319	2.251	2.147	2.204	2.242	2.211	2.228	2.271	2.327	2.343	2.372	
	Ebn at Noon	731	798	822	817	837	838	821	803	786	745	680	683	
	Edh at Noon	71	91	114	140	138	134	136	128	112	90	72	62	
All-Sky Solar Radiation	RadAvg	0.97	1.77	2.96	4.41	5.25	5.72	5.51	4.94	3.52	2.15	1.10	0.76	
	RadStd	0.13	0.26	0.45	0.45	0.59	0.46	0.51	0.48	0.46	0.25	0.13	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	+0.71	N/A	N/A	+1.28	N/A	N/A	-126	N/A	+111	+66
(47)	Regional (76 neighbors)	+0.59	N/A	N/A	+0.85	+0.49	+0.43	-128	-177	+89	+41

Nomenclature: See separate page